

Visualization Draft

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1. Set up

Packages

Data

2. Project background

Dissolved oxygen is an important water quality measurement because it affects which organisms can live in aquatic habitats. Fish, aquatic invertebrates, microbes, and other organisms depend on oxygen availability, so low dissolved oxygen can limit habitat quality. Dissolved oxygen can also influence nutrient cycling because oxygen conditions affect microbial processes that control how nutrients move through wetland and aquatic systems.

This project focuses on two questions:

1. How does dissolved oxygen differ across sites?
2. How does dissolved oxygen stratification differ across sites?

At NCOS, dissolved oxygen may differ across sites because location within the restored wetland can affect water movement, mixing, vegetation, algae, and water level. East Channel does not have the same direct water inputs as the other sites and has more pickleweed nearby. Phelps Bridge is closer to residential areas and has more vegetation in the water. Venoco Bridge is closer to the ocean and may be more influenced by tidal connection or coastal water movement.

These site differences matter because vegetation and algal growth can influence oxygen levels. Photosynthesis can add oxygen during the day, while respiration and decomposition can reduce oxygen. Because each site sits in a different part of NCOS, dissolved oxygen may vary strongly by location.

For this analysis, elevation code is used as a simple way to compare dissolved oxygen at different positions in the water column: bottom, middle, and upper measurements. Larger differences between bottom and upper measurements suggest stronger dissolved oxygen stratification. Stratification is relevant to the physical processes in this study because it reflects how dissolved oxygen conditions can differ vertically when water is not fully mixed.

3. Clean data

Cleaning metadata

Goal

Clean the metadata dataset so it includes survey date, site information, water year, and readable site names.

Cleaning water parameter data

Goal

Clean the water quality parameter dataset so it includes elevation code, elevation labels, and dissolved oxygen measurements.

4. Join data frames

Water quality = metadata + water parameters

5. Filter data for dissolved oxygen analysis

Goal

Create a filtered dissolved oxygen dataset that focuses on the three main NCOS sites, the 2024 and 2025 water years, and the main elevation codes used for the stratification question.

6. Summarize dissolved oxygen

These summaries answer the two main project questions:

1. How does dissolved oxygen differ across sites?
2. How does dissolved oxygen stratification differ across sites?

Dissolved oxygen by site

Table 1: Dissolved oxygen summary by site

Site	Mean DO	Median DO	Q1 DO	Q3 DO	SD	Variance	n
East Channel	7.22	7.28	5.44	9.65	3.43	11.76	109
Phelps Bridge	4.30	3.70	2.56	5.96	2.51	6.33	166
Venoco Bridge	5.68	5.98	2.29	8.35	4.01	16.09	257

Dissolved oxygen by site and elevation code

Table 2: Dissolved oxygen summary by site and elevation code

Site	Elevation code	Mean DO	Median DO	Q1 DO	Q3 DO	SD	n
East Channel	0	7.02	6.66	4.98	9.79	3.62	54
East Channel	1	7.53	7.96	6.15	9.64	3.20	46
East Channel	2	6.86	7.17	5.47	9.12	3.64	9
Phelps Bridge	0	4.53	4.14	2.94	6.05	2.34	76
Phelps Bridge	1	4.19	3.63	2.22	5.90	2.61	81
Phelps Bridge	2	3.31	2.01	0.67	5.94	3.08	9
Venoco Bridge	0	4.01	4.31	0.27	6.56	3.40	89
Venoco Bridge	1	6.00	6.38	2.66	8.80	4.16	99
Venoco Bridge	2	7.38	7.62	5.50	9.10	3.72	69

Stratification difference by site

For this draft, stratification difference is calculated as:

$$\text{Upper median DO} - \text{Bottom median DO}$$

Positive values mean that median DO was higher at the upper measurement position than at the bottom measurement position.

Table 3: Dissolved oxygen stratification difference by site

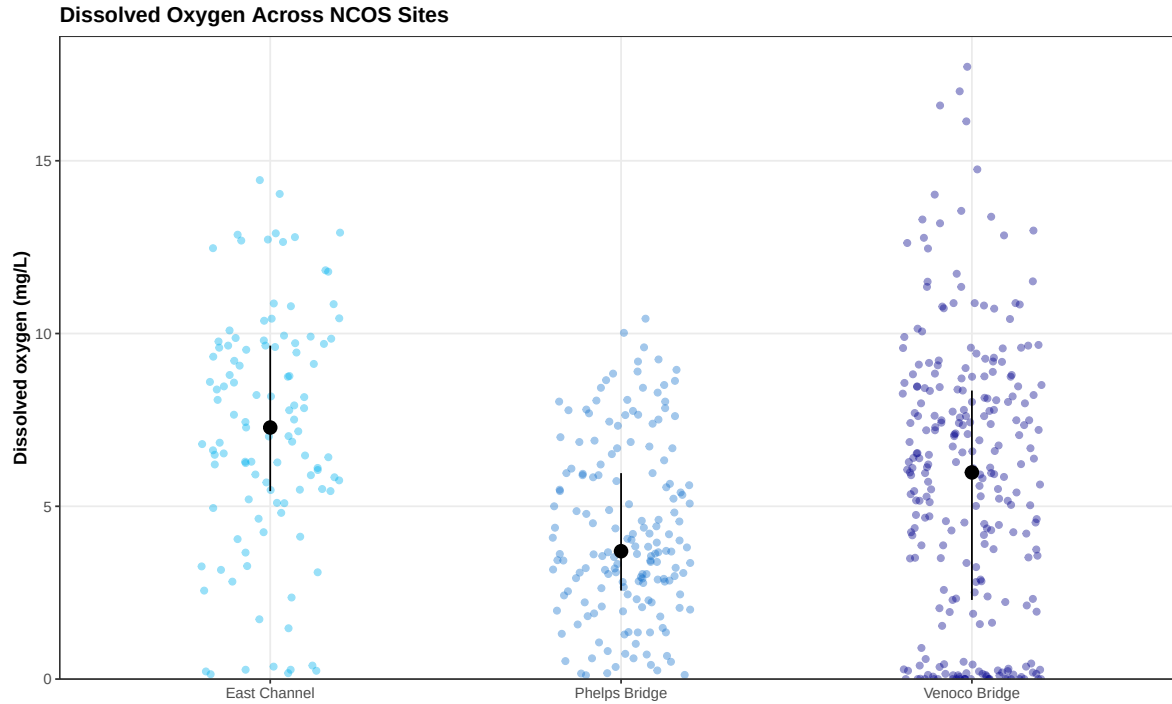
Site	Bottom median DO	Upper median DO	Upper - bottom median DO
East Channel	6.66	7.17	0.50
Phelps Bridge	4.14	2.01	-2.13
Venoco Bridge	4.31	7.62	3.31

7. Visualizing

These figures focus on dissolved oxygen. Figure 1 compares dissolved oxygen across sites. Figure 2 compares dissolved oxygen across elevation codes within each site. Figure 3 uses the recommended elevation-profile style to show median dissolved oxygen across elevation codes. Figure 4 adds an exploratory time component by showing median dissolved oxygen through time by elevation code and site.

Figure 1: Dissolved oxygen across sites

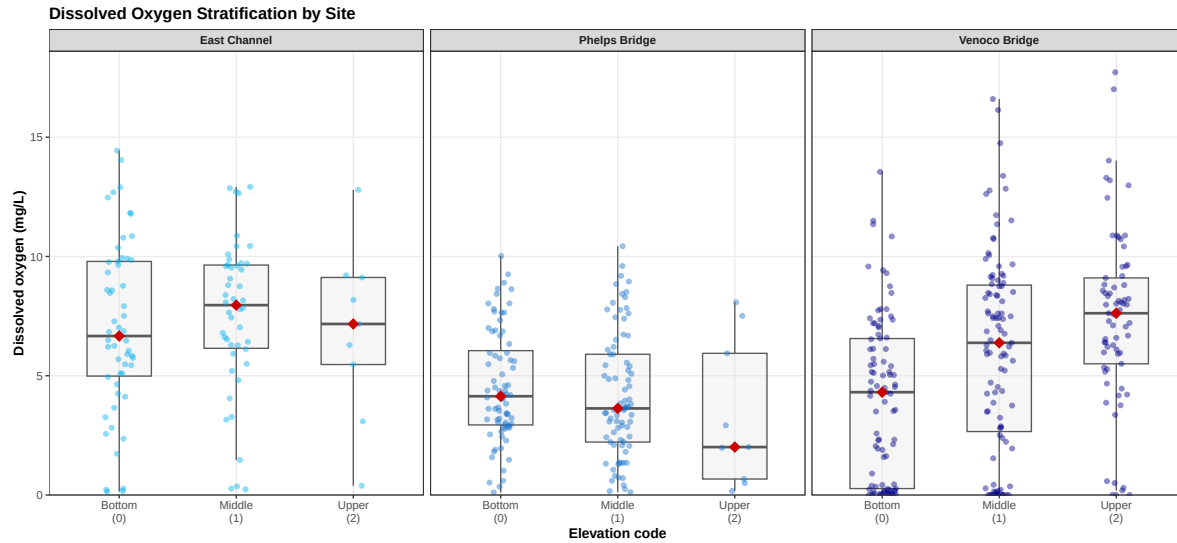
Caption: Figure 1 shows dissolved oxygen measurements across the three main NCOS sites. Points represent individual observations. Larger black points represent site medians, and vertical ranges show the 1st to 3rd quartile range.



Interpretation: Dissolved oxygen varied by site. East Channel had the highest median DO, Phelps Bridge had the lowest median DO, and Venoco Bridge had the widest spread of values.

Figure 2: Dissolved oxygen stratification by site

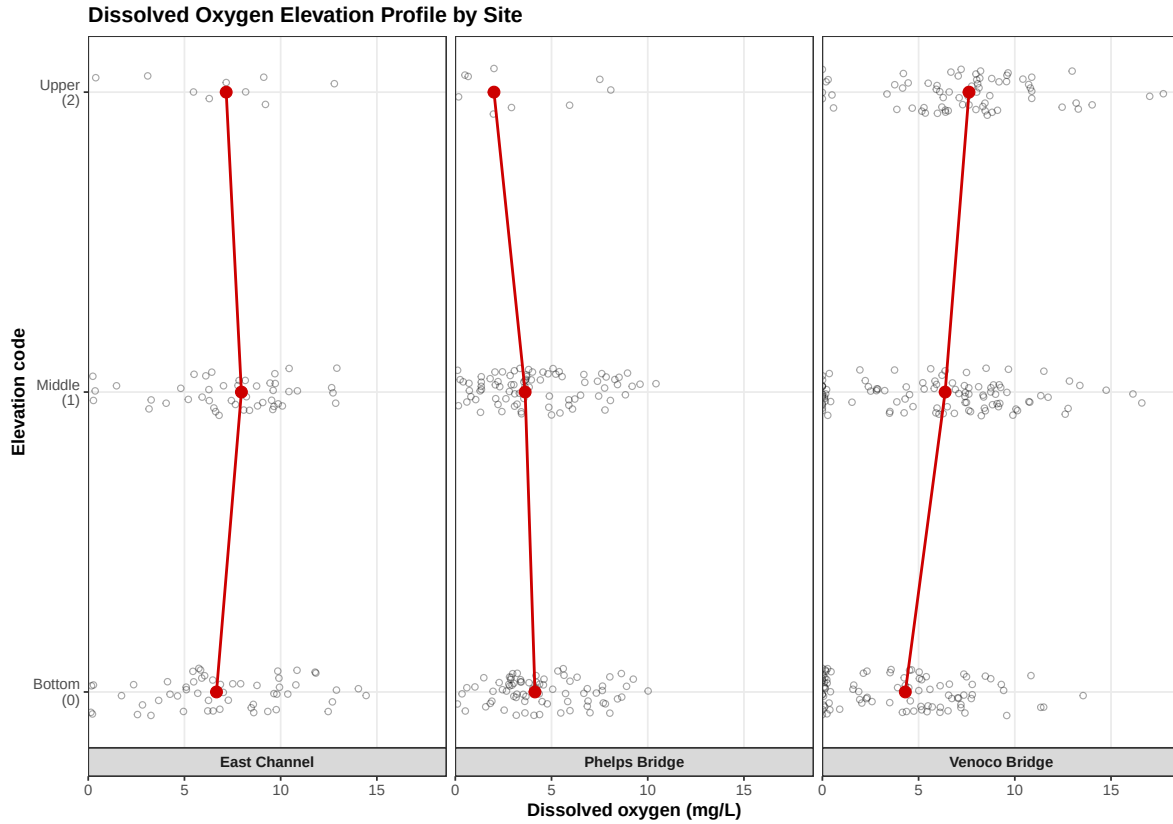
Caption: Figure 2 shows dissolved oxygen measurements across elevation-code positions within each site. Boxplots show the distribution of DO values, points represent individual observations, and red diamonds represent median DO values.



Interpretation: DO patterns across elevation codes were different by site. East Channel was fairly similar across depths, Phelps Bridge stayed lower overall, and Venoco Bridge showed the clearest increase from bottom to upper measurements.

Figure 3: Dissolved oxygen elevation profile by site

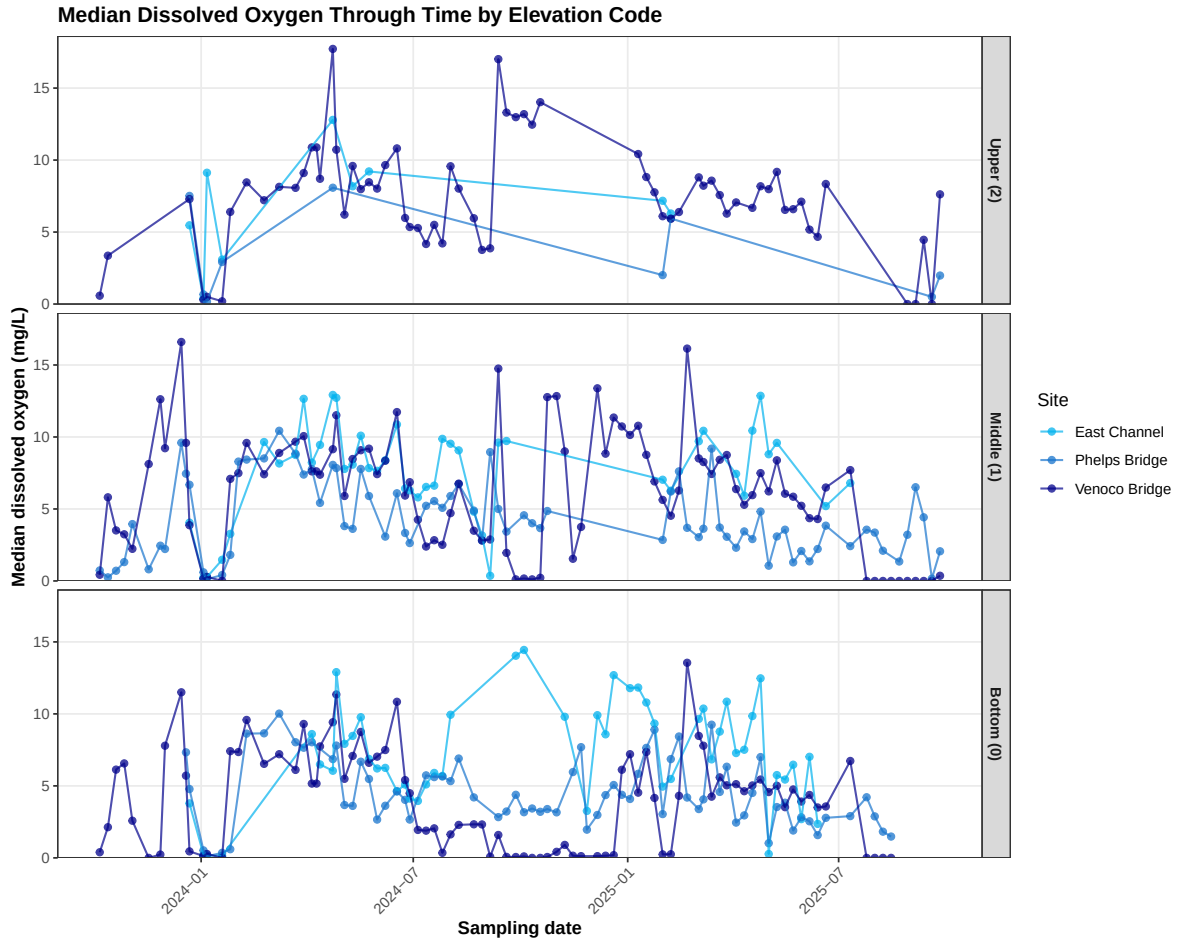
Caption: Figure 3 shows dissolved oxygen elevation profiles for each site. Points represent individual DO observations, red points represent median DO values, and red lines connect median DO values across elevation codes within each site.



Interpretation: Venoco Bridge had the clearest elevation profile, with median DO increasing from bottom to upper measurements. East Channel and Phelps Bridge showed smaller differences across elevation codes.

Figure 4: Median dissolved oxygen through time by elevation code

Caption: Figure 4 shows median dissolved oxygen through time, separated by elevation code. Points represent median DO for each sampling date and site, and lines connect values within each site.



Interpretation: Median DO changed through time, but the site pattern was still visible. East Channel was generally higher, Phelps Bridge was generally lower, and Venoco Bridge showed the clearest difference between bottom and upper measurements.

8. Statistical analysis

The statistical analyses follow the class workflow for comparing more than two groups: summarize the data, visualize the pattern, run a Kruskal-Wallis test, use Dunn post-hoc tests when the Kruskal-Wallis result is significant, calculate effect size, and interpret the result. The main response variable is dissolved oxygen. The grouping variables are site and elevation code.

Question 1: How does dissolved oxygen differ across sites?

This analysis tests whether dissolved oxygen differs across East Channel, Phelps Bridge, and Venoco Bridge.

Kruskal-Wallis test

```
Kruskal-Wallis rank sum test
```

```
data: do_mg_l by site_full
```

```
Kruskal-Wallis chi-squared = 45.667, df = 2, p-value = 1.212e-10
```

The Kruskal-Wallis test showed that dissolved oxygen differed significantly across sites, $\chi^2 = 45.67$, $df = 2$, $p < 0.001$. This means that DO was not distributed the same way across East Channel, Phelps Bridge, and Venoco Bridge.

Dunn post-hoc test

```
# A tibble: 3 x 9
```

.y.	group1	group2	n1	n2	statistic	p	p.adj	p.adj.signif
* <chr>	<chr>	<chr>	<int>	<int>	<dbl>	<dbl>	<dbl>	<chr>
1 do_mg_l	East Chan~	Phelp~	109	166	-6.74	1.57e-11	4.72e-11	****
2 do_mg_l	East Chan~	Venoc~	109	257	-4.03	5.62e- 5	1.12e- 4	***
3 do_mg_l	Phelps Br~	Venoc~	166	257	3.72	1.98e- 4	1.98e- 4	***

Because the Kruskal-Wallis test was significant, Dunn post-hoc tests were used to compare site pairs. All three site pairs were significantly different after adjustment: East Channel and Phelps Bridge, East Channel and Venoco Bridge, and Phelps Bridge and Venoco Bridge.

Effect size

```
# A tibble: 1 x 5
```

.y.	n	effsize	method	magnitude
* <chr>	<int>	<dbl>	<chr>	<ord>
1 do_mg_l	532	0.0825	eta2[H]	moderate

The effect size for the site comparison was $\eta^2[H] = 0.083$, which was interpreted as moderate in the test output. This suggests that site explained a moderate amount of variation in DO ranks in this filtered dataset.

Interpretation

Dissolved oxygen differed across the three main NCOS sites. East Channel had the highest median DO, Phelps Bridge had the lowest median DO, and Venoco Bridge had the widest spread of values. The Dunn post-hoc test showed that all site pairs were significantly different, so DO patterns were not the same across sites.

Question 2: How does dissolved oxygen stratification differ across sites?

This analysis tests whether dissolved oxygen differs across elevation-code measurement positions within each site. This addresses stratification because elevation code represents the measurement position in the water column.

Kruskal-Wallis tests within each site

```
Kruskal-Wallis rank sum test
```

```
data: do_mg_l by elevation_code
```

```
Kruskal-Wallis chi-squared = 1.1119, df = 2, p-value = 0.5735
```

```
Kruskal-Wallis rank sum test
```

```
data: do_mg_l by elevation_code
```

```
Kruskal-Wallis chi-squared = 3.0201, df = 2, p-value = 0.2209
```

```
Kruskal-Wallis rank sum test
```

```
data: do_mg_l by elevation_code
```

```
Kruskal-Wallis chi-squared = 28.641, df = 2, p-value = 6.034e-07
```

The within-site Kruskal-Wallis tests showed no significant DO differences across elevation codes at East Channel or Phelps Bridge. Venoco Bridge did show a significant difference across elevation codes, $\chi^2 = 28.64$, $df = 2$, $p < 0.001$. This means that Venoco Bridge had the clearest statistical evidence of DO stratification.

Dunn post-hoc test for Venoco Bridge

```
# A tibble: 3 x 9
  .y.    group1 group2    n1    n2 statistic      p      p.adj p.adj.signif
* <chr>  <chr>  <chr>  <int> <int>    <dbl>    <dbl>    <dbl> <chr>
1 do_mg_l 0      1      89    99      3.39 0.000688    1.38e-3 **
2 do_mg_l 0      2      89    69      5.26 0.000000145 4.36e-7 ****
3 do_mg_l 1      2      99    69      2.22 0.0266     2.66e-2 *
```

Because Venoco Bridge had a significant Kruskal-Wallis result, Dunn post-hoc tests were used to compare elevation-code pairs at that site. All three pairs were significantly different after adjustment: bottom and middle, bottom and upper, and middle and upper. This supports the pattern of higher median DO from bottom to upper measurements at Venoco Bridge.

Effect size for Venoco Bridge

```
# A tibble: 1 x 5
  .y.      n effsize method magnitude
* <chr>  <int>  <dbl> <chr>    <ord>
1 do_mg_l 257  0.105 eta2[H] moderate
```

The effect size for the Venoco Bridge elevation-code comparison was $\eta^2[H] = 0.105$, which was interpreted as moderate in the test output. This suggests that elevation code explained a moderate amount of variation in DO ranks at Venoco Bridge.

Interpretation

Statistical evidence for DO stratification was clearest at Venoco Bridge. East Channel and Phelps Bridge did not show significant DO differences across elevation codes, while Venoco Bridge did. At Venoco Bridge, median DO was lower at the bottom and higher at the middle and upper positions.

9. Advanced elective plan

For the advanced elective, the group plans to create a physical version of a dissolved oxygen stratification model. We will use three clear containers to represent East Channel, Phelps Bridge, and Venoco Bridge. Each container will show bottom, middle, and upper elevation-code positions.

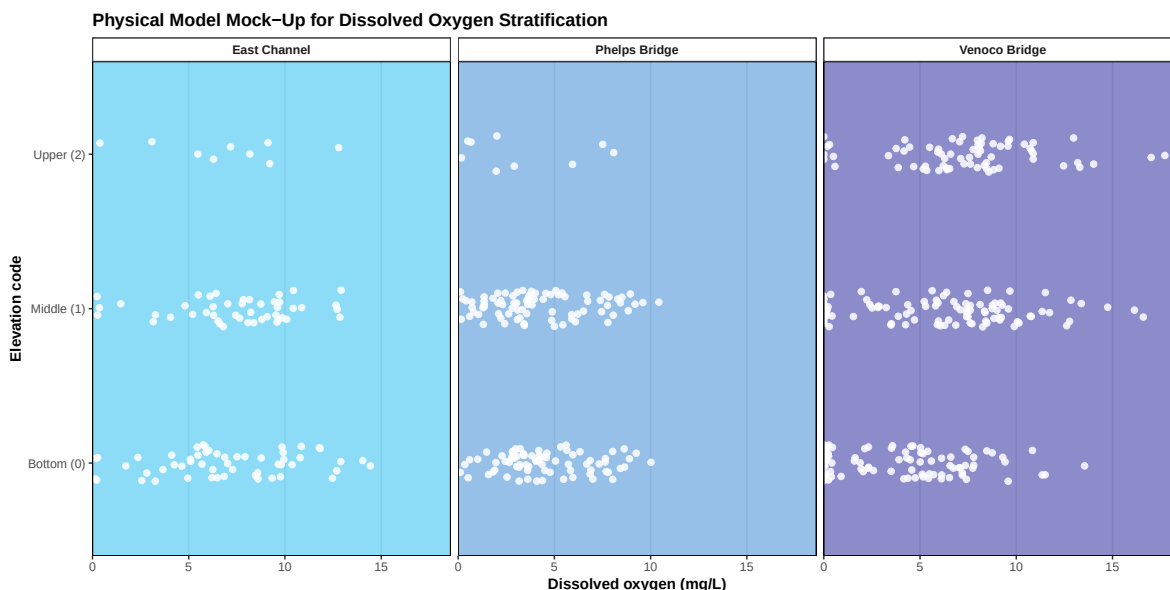
Blue beads will represent water in each container. The beads have already been separated into different shades of blue so each site can be visually distinct. Cotton from cotton balls will be layered more densely or sparsely to represent relative dissolved oxygen concentration at each elevation position. The visual mock-up below was created to help guide the physical model.

Elective schedule

By Week 9, we separated the blue beads by color, created the visual mock-up, and finalized the general plan for representing the three sites.

During Week 10, we will finish the final layering plan for each tube, add markings to show bottom, middle, and upper elevation-code positions, and assemble the physical model.

By the final presentation, the three containers will be completed and used to explain how dissolved oxygen stratification differs across the NCOS sites.



The physical model will use the same basic idea as the mock-up: each site will have a visible water-column structure, and the bottom, middle, and upper positions will be represented separately. Based on the data, Venoco Bridge should show the clearest difference between bottom and upper measurements, while East Channel and Phelps Bridge should show smaller vertical differences.

10. Overall interpretation

Overall, dissolved oxygen differed across the three main NCOS sites, and statistical evidence for stratification was clearest at Venoco Bridge. East Channel had the highest median DO, Phelps Bridge had the lowest median DO, and Venoco Bridge showed the clearest increase from bottom to upper measurements. These results suggest that DO conditions and stratification patterns were not the same across the three sites.